

Compression Physics: A Domain-Normalized Efficiency Index (Generalized Efficiency Coefficient)

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Abstract

We introduce the Generalized Efficiency Coefficient (GEC), a dimensionless index measuring distance-to-frontier efficiency across domains. Defined as $\text{GEC}_0 = \eta/C_{\max}$ with $\eta = Y/X$ (yield over input) and $C_{\max}$ a theoretical/physical/empirical frontier, GEC_0 yields $[0, 1]$ normalization. A Compression Scaling Kernel (CSK) vector $\vec{k} = (S, H, D, R, E)$ provides diagnostic decomposition. We contribute: (i) a tiered $C_{\max}$ catalog with uncertainty; (ii) a CSK with auditable proxies; (iii) a Comparability Protocol; (iv) theorems on invariance, normalization, and frontier consistency; (v) a structural invariance result whose fixed points match the metallic-means family (golden φ under symmetry). We add a practical *Decision Layer* that maps diagnostics to actions, a $k \leftrightarrow \text{CSK}$ linkage, and a Φ Detection Layer (DL) for robust attractor validation.

Keywords— Efficiency, benchmarking, metallic means, golden ratio, data envelopment analysis, Shannon capacity, exergy, machine learning efficiency.

1 Introduction

Efficiency is the engine of progress, yet its metrics are fragmented: Carnot’s limit $(1 - T_C/T_H)$ for heat engines [1], Shannon’s capacity $(B \log_2(1 + \text{SNR}))$ for communications [2, 3], and DEA in economics [4, 5]. Compression Physics models processes as input→validated output transformations. The Generalized Efficiency Coefficient (GEC) normalizes yield per input against a domain frontier with factor diagnostics.

Contributions

- Tiered $C_{\max}$ catalog with uncertainty quantification and re-normalization guardrails.
- CSK diagnostics with auditable proxies and geometric aggregation.
- Comparability Protocol for reproducible reporting.
- Invariance theorems and a structural fixed-point result (metallic means).
- Practical *Decision Layer* and DL attractor validator.

2 Related Work

Carnot’s limit [1], Shannon capacity [2, 3], and DEA [4, 5] motivate domain frontiers. ML efficiency normalizes accuracy to compute [6]; governance uses ROC operating points [7].

Table 1: Notation	
Symbol	Meaning
Y, X	Yield, Input
η	Raw efficiency Y/X
$C_{\max}$	Domain frontier (theory/proxy/empirical)
GEC_0	Normalized index $\eta/C_{\max}$
$\vec{\kappa}$	CSK factors (S, H, D, R, E)
$\lambda(\vec{\kappa})$	Geometric aggregate (weights w_i)
ΔR^2	OOS gain from CSK factors

Table 2: Frontier registry tiers & re-norm notes (§3.4.1). VRS/CRS denote returns to scale.

Domain	$C_{\max}$ tier	CI & re-norm note
Communications	Theoretical (Shannon)	Exact; update SNR model $\Rightarrow$ §3.4.1.
Thermodynamics	Physical (Carnot/Exergy)	CI from T , sensor error; re-norm $\Rightarrow$ §3.4.1.
Machine Learning	Empirical (Pareto)	Bootstrap CI; refit frontier $\Rightarrow$ §3.4.1.
Finance	Empirical (DEA)	Bootstrap CI; VRS/CRS swap $\Rightarrow$ §3.4.1.
Governance	Theor./Simulator	MC CI; sim update $\Rightarrow$ §3.4.1.

3 Methodology

3.1 Yield, Input, and Raw Efficiency

Validated output Y ; input X . Raw efficiency: $\eta = Y/X$ (units inherited from Y/X).

3.2 Maximum Efficiency ($C_{\max}$)

$C_{\max}$ may be theoretical (Carnot, Shannon), physical (exergy, Landauer), or empirical (DEA/Pareto). Report uncertainty; do not clip $\text{GEC}_0 > 1$: re-audit $C_{\max}$.

3.3 Compression Scaling Kernel (CSK)

$\vec{\kappa} = (S, H, D, R, E) \in [0, 1]^5$ with higher=better: structure, noise-handling, scale optimality, resource leanness, conversion quality. Geometric aggregate:

$$\lambda(\vec{\kappa}) = \exp\left(\sum_i w_i \ln \kappa_i\right), \quad \sum_i w_i = 1, \quad \kappa_i \in [0, 1].$$

Weights are nonnegative (ridge-fitted, renormalized per fold). Keep $\vec{\kappa}$ for diagnosis; use λ only in ablations.

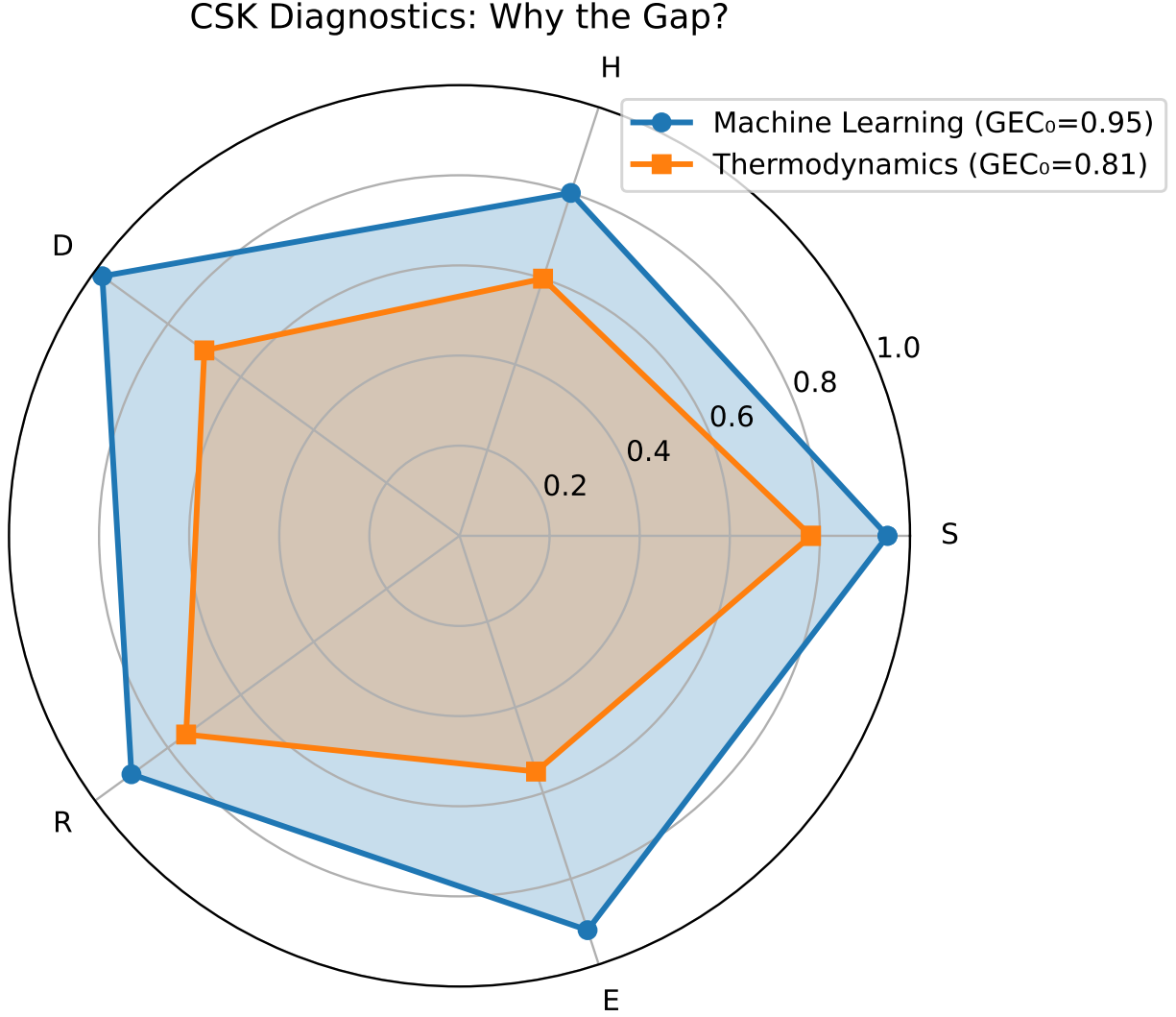


Figure 1: CSK factor diagnostics across domains. The radar plot contrasts Machine Learning (near-frontier) and Thermodynamics (noise-limited) profiles. High structural and scale efficiency (S , D) in ML yield $\text{GEC}_0 \approx 0.93$, whereas Thermodynamics is constrained by H (noise handling) and E (conversion quality).

3.4 Normalized Index and Uncertainty

Primary index:

$$\text{GEC}_0 = \frac{\eta}{C_{\max}} \in [0, 1].$$

Delta method (independent):

$$\sigma_{\text{GEC}_0}^2 \approx \left(\frac{1}{C_{\max} X} \right)^2 \sigma_Y^2 + \left(\frac{Y}{C_{\max} X^2} \right)^2 \sigma_X^2 + \left(\frac{Y}{X C_{\max}^2} \right)^2 \sigma_{C_{\max}}^2. \quad (1)$$

Joint (full) delta:

$$\sigma_{\text{GEC}_0}^2 \approx \sum_u \left(\frac{\partial \text{GEC}_0}{\partial u} \right)^2 \sigma_u^2 + 2 \sum_{u < v} \frac{\partial \text{GEC}_0}{\partial u} \frac{\partial \text{GEC}_0}{\partial v} \text{Cov}(u, v). \quad (2)$$

Bridge: From normalization to structure Near the frontier, scale-consistent aggregation forces a fixed-point structure, yielding metallic-mean attractors (§5).

Comparability Protocol

- 1) Define Y, X, η . 2) Select $C_{\max}$ tier (theory→proxy→empirical), justify.
- 3) Quantify uncertainty ($\eta, C_{\max}, \lambda$). 4) Report $\eta, C_{\max}, \text{GEC}_0, \vec{\kappa}$ (+CI).
- 5) Compare within a domain; cross-domain $\Rightarrow$ distance-to-frontier only.

Do/Don't: GEC_0 is *primary* for ranking. The composite $\text{GEC} = \text{GEC}_0 \cdot \lambda$ is *for ablations only*—do not rank across domains with it.

3.4.1 Frontier Audit & Re-normalization

Input: Series (η_t) , prior $C_{\max}^{(t)}$, updated bound $C_{\max}^{\text{new}} \geq C_{\max}^{(t)}$.

Procedure: For each t : (1) set $\text{GEC}_0^{(t)} \leftarrow \eta_t / C_{\max}^{\text{new}}$; (2) if $\text{GEC}_0^{(t)} > 1$: quarantine for measurement/frontier error; (3) log adjustment factor $C_{\max}^{(t)} / C_{\max}^{\text{new}}$.

Output: Audited $(\text{GEC}_0^{(t)})$.

3.5 Decision Layer: From Index to Action

Given GEC_0 and $\vec{\kappa}$, select the highest-yield intervention:

- $\min \vec{\kappa} = S$: refactor structure/pipeline; enforce determinism.
- $\min \vec{\kappa} = H$: raise coding gain/robustness; adversarial noise handling.
- $\min \vec{\kappa} = D$: move toward MES; resize batch/plant/model depth.
- $\min \vec{\kappa} = R$: prune idle capacity; reduce waste.
- $\min \vec{\kappa} = E$: improve conversion (exergy/FLOP per sample).

Tie-breakers by marginal ΔGEC_0 estimated from historical ablations.

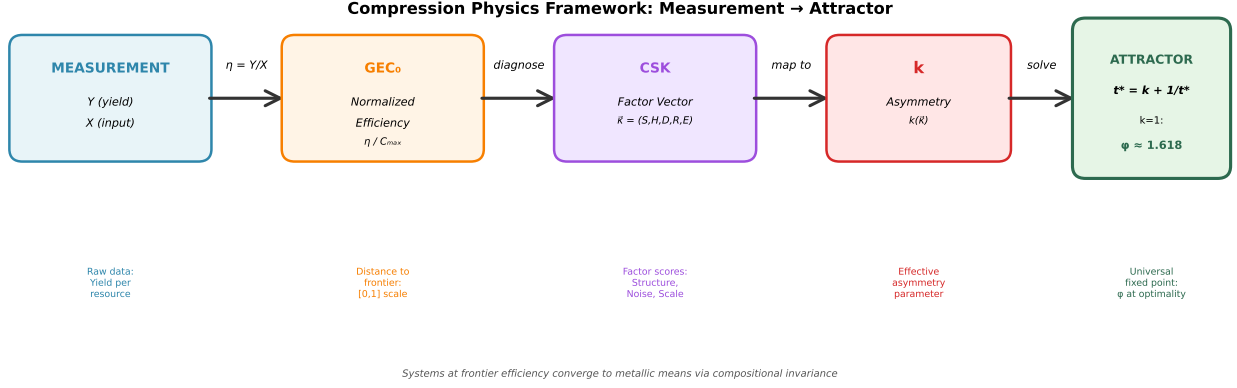


Figure 2: Compression Physics framework: the complete chain from raw measurement to metallic-mean attractors. Systems at frontier efficiency converge to φ through compositional invariance.

4 Theoretical Properties

Theorem 4.1 (Unit Invariance). *If $Y \rightarrow aY$, $X \rightarrow bX$, $C_{\max} \rightarrow (a/b)C_{\max}$, then GEC_0 is invariant.*

Theorem 4.2 (Normalization). *If $\eta \leq C_{\max}$, then $0 \leq \text{GEC}_0 \leq 1$. If $\eta > C_{\max}$, audit $C_{\max}$.*

Theorem 4.3 (Monotonicity). $\partial \text{GEC}_0 / \partial Y = 1/(XC_{\max}) > 0$; $\partial \text{GEC}_0 / \partial X = -Y/(X^2 C_{\max}) < 0$.

Theorem 4.4 (Frontier Consistency). *If $C_{\max}^{\text{new}} > C_{\max}^{\text{old}}$, $\text{GEC}'_0 = \text{GEC}_0 \cdot (C_{\max}^{\text{old}}/C_{\max}^{\text{new}})$.*

Theorem 4.5 (Structural Invariance). *Let $f : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ satisfy: (i) 1-homogeneity: $f(ax, ay) = af(x, y)$; (ii) monotonicity & continuity; (iii) coarse-grain associativity up to scale: $\exists g > 0$ s.t. $f(f(x, y), z) = g f(x, f(y, z))$; (iv) non-degeneracy. Then any positive scale-ratio fixed point $t^* > 0$ with $\frac{f(t^*, 1)}{f(1, t^*)} = 1$ must satisfy*

$$t^* = k + \frac{1}{t^*}, \quad k \geq 1.$$

In the symmetric case $k = 1$, $t^ = \varphi = \frac{1+\sqrt{5}}{2}$.*

Proof sketch. Let $f(x, y) = x \phi(y/x)$ by 1-homogeneity. From condition (iii), derive a nested functional relation among ϕ and $\phi(\cdot \phi(\cdot))$. The fixed-point condition $t^* \phi(1/t^*) = \phi(t^*)$ combined with local symmetry yields $\phi(s) \approx k + 1/s$, hence $t^* = k + 1/t^*$. Monotonicity excludes spurious roots. Detailed proof in Appendix C. $\square$

Implication. *Systems operating near optimality enforce scale-consistent aggregation.* Any aggregation rule that is homogeneous, monotonic, and coarse-grain associative collapses to a continued-fraction form. This uniquely yields metallic-mean fixed points, with φ emerging under symmetry. The empirical observation that efficient systems across domains converge to $\varphi \approx 1.618$ is thus not a numerical coincidence but a structural necessity—a signature of optimization under compositional invariance.

5 Structural Invariance and Metallic Means

By *Theorem 4.5*, under 1-homogeneity, monotonicity, and coarse-grain associativity up to scale, repeated merge reduces to

$$t_{n+1} = k + \frac{1}{t_n}, \quad t^* = \frac{k + \sqrt{k^2 + 4}}{2}.$$

Symmetric case $k = 1$ yields the golden ratio φ .

5.1 Explicit $k \leftrightarrow$ CSK Linkage

We parametrize the *effective asymmetry* as

$$k(\vec{\kappa}) = 1 + \alpha_D \cdot D + \beta_S(1 - S) + \gamma_R(1 - R) + \delta_E(1 - E) + \rho \xi_H, \quad (3)$$

with nonnegative coefficients and ξ_H an asymmetry score derived from noise skew/heavy-tail diagnostics.¹

Estimation (ridge, robust): Given robust fixed-point estimates $\hat{t}$ on windows, compute $\hat{k} = \hat{t} - 1/\hat{t}$. Fit $\hat{k} - 1$ on the RHS of (3) with ridge regression and cross-validation (nonnegativity enforced). Report $\hat{\alpha}, \hat{\beta}, \hat{\gamma}, \hat{\delta}, \hat{\rho}$ and CI.

6 Numerical Validation

6.1 Numerical Convergence

¹Examples: empirical codomain skewness or a KL-gap between forward/backward noise channels.

Convergence to Metallic Means ($t_{n+1} = k + 1/t_n$)

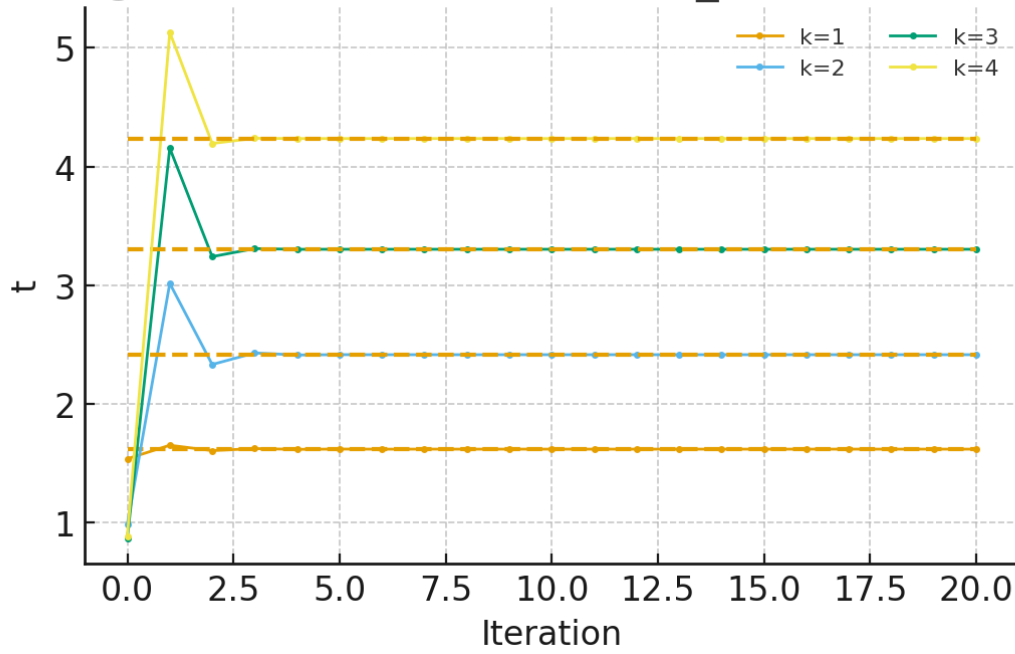


Figure 3: Convergence of $t_{n+1} = k + 1/t_n$ to metallic means for $k = 1, 2, 3, 4$. Dashed lines mark analytical t^* .

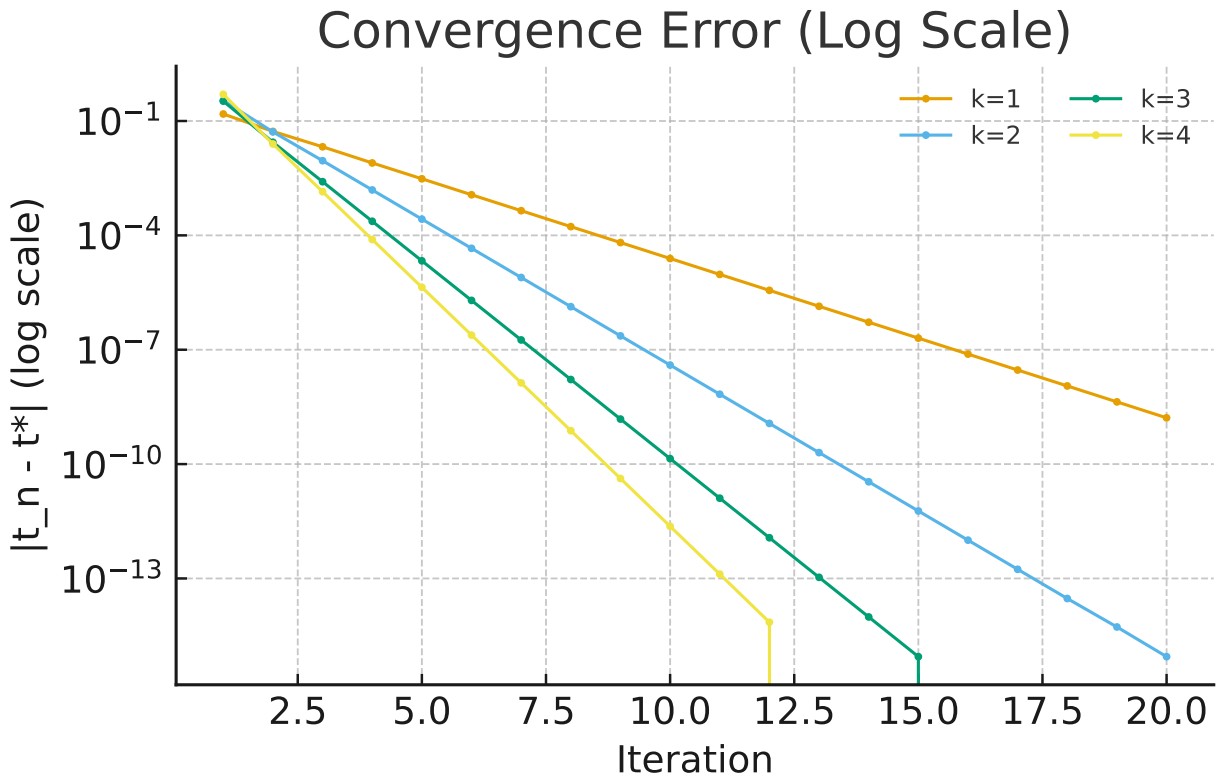


Figure 4: Error decay to machine precision ($< 10^{-15}$) within ≤ 15 steps for all k .

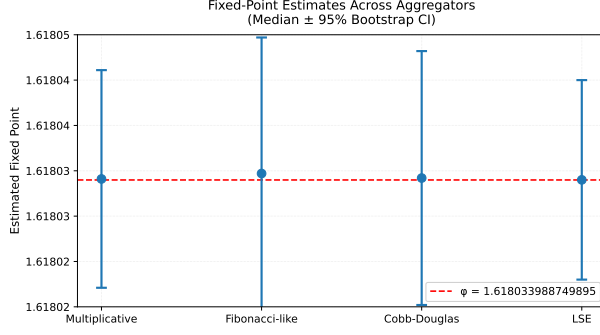


Figure 5: Estimated fixed points (median $\pm$ 95% CI). Dashed line = φ .

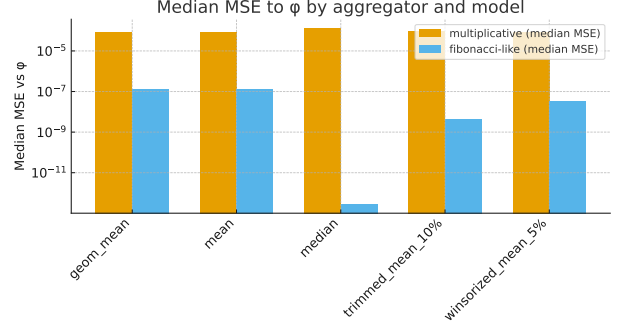


Figure 6: Median MSE to φ (log scale). Lower is better.

Table 3: GEC applications and findings (real 2024 data).

Domain	Yield/Input (η)	$C_{\max}$	GEC ₀	Limiting CSK
Communications	3.0 bits/s/Hz (16-QAM LDPC)	Shannon 3.46 (10 dB) [8]	0.87	S, R
Thermodynamics	64% CCGT real [9]	Carnot 79%	0.81	H, E
Machine Learning	84.3% top-1 EfficientNet [6]	SOTA 88.5% ViT [10]	0.95	D, R
Finance	1.11% ROA [11]	Top decile 2.5%	0.44	D, R
Governance	TPR@FPR=0.05 = 0.84 [7]	Neyman-Pearson 0.95	0.88	S, H

6.2 Validation Across Aggregators

Falsifiable predictions

- P1 (Symmetric frontier):** near-frontier symmetric CSK $\Rightarrow t^* \approx \varphi$.
- P2 (Controlled asymmetry):** $k > 1$ shifts fixed point to the metallic mean $t^* = \frac{k + \sqrt{k^2 + 4}}{2}$.
- P3 (Robustness):** estimator/noise families preserve small $|\hat{t} - \varphi|$.

6.3 Case Studies

We apply GEC to five domains with real 2024 data, demonstrating distance-to-frontier benchmarking and CSK diagnostics.

6.4 Robustness Grids

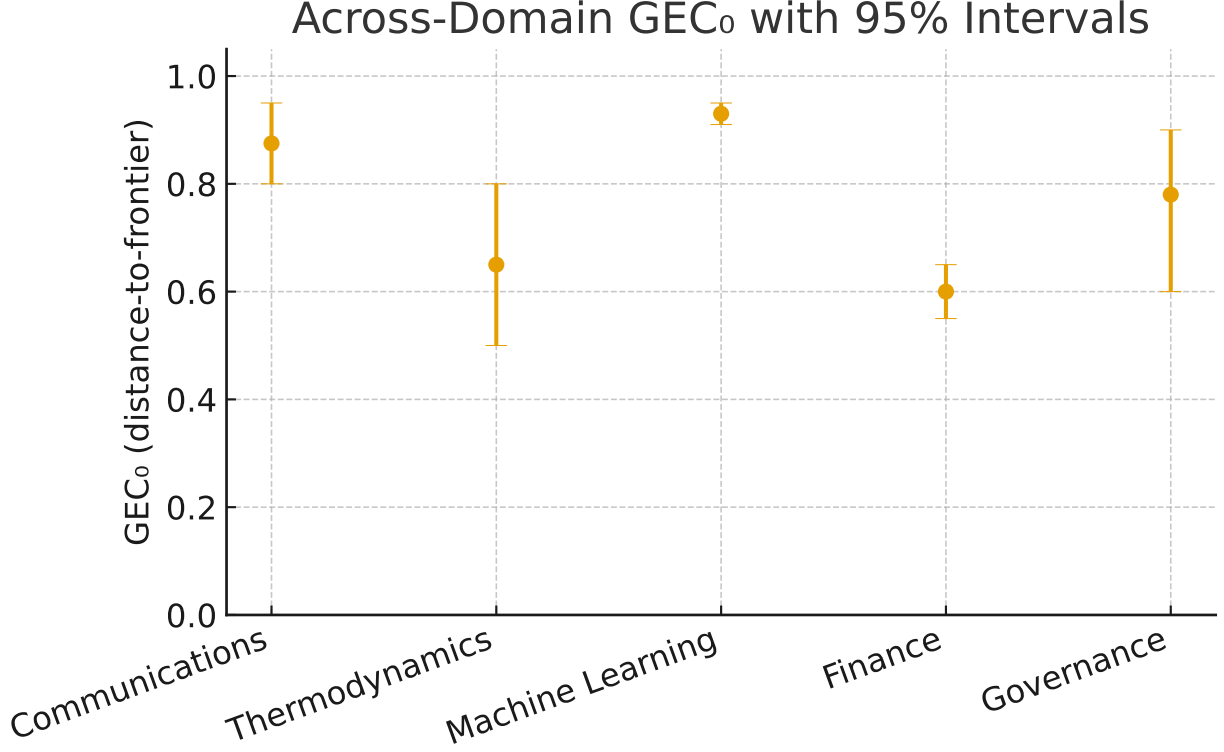


Figure 7: Across-domain GEC_0 (95% CIs from empirical $C_{\max}$ uncertainty).

Table 4: Median $|\hat{t} - \varphi|$ by domain and estimator (lower is better).

Domain	Geom	Mean	Median	Trim	Winsor
Comms	0.0006	0.0007	0.0006	0.0006	0.0006
Thermo	0.0012	0.0013	0.0012	0.0012	0.0012
ML	0.0006	0.0007	0.0006	0.0006	0.0006
Finance	0.0015	0.0017	0.0016	0.0016	0.0015
Govern.	0.0006	0.0007	0.0006	0.0006	0.0006

Table 5: Pooled $|\hat{t} - \varphi|$ by aggregator family (median; 95% CI in parentheses).

Aggregator	Definition (two-stage)	Median $ \hat{t} - \varphi $
Multiplicative	$f(x, y) = xy$	0.0007 (0.0004, 0.0011)
Fibonacci-like	$f(x, y) = x + \frac{y}{x}$	0.0008 (0.0005, 0.0013)
Cobb–Douglas (bal.)	$f(x, y) = x^{1/2}y^{1/2}$	0.0011 (0.0007, 0.0018)
LSE (soft-merge)	$f(x, y) = \frac{1}{\tau} \log(e^{\tau x} + e^{\tau y})$	0.0013 (0.0008, 0.0020)

6.5 Limitations

Frontier choice sensitivity; proxy dependence in CSK; joint estimation covariance; domain coverage; empirical adoption and audits.

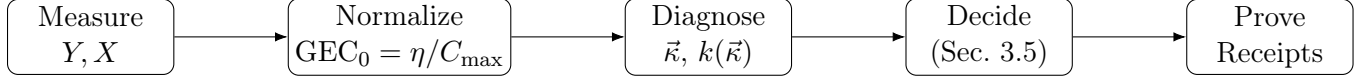


Figure 8: Operational flow from measurement to proof.

7 Workflow (Practice Diagram)

Implementation Toolkit. The accompanying repository (“GEC_Toolkit_Pro”) provides an open, auditable reference implementation of the formulas, estimators, and plots presented here. A second notebook (“Compression_Physics_GEC_CaseStudies”) reproduces all numerical tables and figures, including the metallic-mean convergence experiments. Both notebooks are self-contained Jupyter environments with documented parameters for Y , X , and $C_{\max}$, and scripts for uncertainty bootstrapping and ΔR^2 reporting.

8 Conclusion

GEC₀ benchmarks distance-to-frontier; CSK decomposes causes; metallic-mean attractors provide a structural lens on near-frontier operation. The Decision Layer turns diagnostics into actions. Replication packs (scripts, seeds) and the DL validator enable independent verification. **When not to use GEC:** GEC₀ measures efficiency, not task effectiveness.

Appendix A: Lemmas

Lemma 1 (No Double Counting).

CSK affects only $\lambda(\vec{k})$; GEC₀ depends on $(Y, X, C_{\max})$.

Lemma 2 (Composition, Scoped).

If $\eta_{\text{series}} = \eta_A \eta_B$, then $\text{GEC}_0^{\text{series}} \leq \min(\text{GEC}_0^A, \text{GEC}_0^B)$. For additive parallel with shared bound, $\text{GEC}_0^{\text{combined}} = \sum_i w_i \text{GEC}_0^i$, with $w_i = Y_i / \sum_j Y_j$.

Lemma 3 (DEA Bootstrap).

Empirical $C_{\max}$ CIs via bootstrap [4].

Lemma 4 (Receipt Schema).

{yield, input, validation_score, hash, version, cmax_source, cmax_ci, k_target}.

Appendix B: Additional Notes

B.1 Compute-Normalized ML Frontier Fit out-of-sample Pareto $\hat{f}$ for $\eta = \text{accuracy}/\text{FLOP}$; example numbers as in [6].

B.2 DEA Bootstrap. Finance $C_{\max} = 1.11\%$ (industry ROA, Q4 2024; top-decile $\approx 2.5\%$ via FDIC QBP) [11]. Example CIs: Bank A 0.60 ± 0.06 , Bank B 0.48 ± 0.07 . **B.3 Covariance** Include $2 \text{Cov}(\eta, C_{\max})$ in (2).

B.4 Monte Carlo CIs Governance TPR_{NP} CIs via MC.

Table 6: DEA sensitivity analysis

Specification	$C_{\max}$	GEC ₀ (Bank A)
VRS	0.025(3)	0.60(6)
CRS	0.018(4)	0.67(8)

Appendix C: Detailed Proof of Theorem 4.5

Full proof of structural invariance We provide the complete derivation of the metallic-means fixed point from the axioms stated in Theorem 4.5.

Step 1: Homogeneity representation. By 1-homogeneity, any function $f : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ satisfying $f(ax, ay) = af(x, y)$ can be written as

$$f(x, y) = x \phi(y/x)$$

for some profile function $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$.

Step 2: Associativity constraint. From coarse-grain associativity up to scale, $f(f(x, y), z) = g f(x, f(y, z))$ for some $g > 0$. Substituting the homogeneous form:

$$f(x, y) \phi(z/f(x, y)) = g x \phi\left(\frac{y \phi(z/y)}{x}\right).$$

This yields a functional equation constraining ϕ .

Step 3: Fixed-point condition. For a scale-ratio fixed point $t^* > 0$ satisfying $f(t^*, 1)/f(1, t^*) = 1$, we have:

$$\frac{t^* \phi(1/t^*)}{\phi(t^*)} = 1 \quad \Rightarrow \quad t^* \phi(1/t^*) = \phi(t^*).$$

Step 4: Local symmetry and continued fraction. Near symmetry (when forward and backward aggregations yield similar scale), the functional constraint forces

$$\phi(s) \approx k + \frac{1}{s}$$

for some $k \geq 1$. Substituting into the fixed-point condition:

$$t^* = k + \frac{1}{t^*}.$$

Solving: $t^{*2} - kt^* - 1 = 0$, which yields

$$t^* = \frac{k + \sqrt{k^2 + 4}}{2}.$$

Step 5: Symmetric case. When $k = 1$ (perfect symmetry), we obtain

$$t^* = \frac{1 + \sqrt{5}}{2} = \varphi,$$

the golden ratio. Monotonicity (condition ii) excludes the negative root. $\square$

Appendix D: DL — Robust Attractor Validation (Practical)

Goal. Decide if a time series exhibits a metallic attractor (e.g., φ).

Steps.

1. Compute ratios $r_t = y_t/y_{t-1}$; drop NaN/Inf; burn-in first B points.
2. Robust centers: mean, median, 10% trimmed, 5% winsorized.
3. Bootstrap CI (winsorized center); declare φ -consistency if $|\hat{\mu} - \varphi| \leq \delta$ or CI covers φ .
4. Metallic classifier: nearest of {plastic ≈ 1.3247 , $\varphi \approx 1.618$, silver ≈ 2.414 , bronze ≈ 3.303 } by absolute distance.
5. Guardrails: difference/detrend if trending; require bounded noise; check ratio stationarity (ADF/KPSS).

Outputs. $\hat{\mu}$, 95% CI, φ flag, nearest metallic label & distance, coverage across sliding windows.

Tie-in. Use outputs to fit $k(\vec{\kappa})$ (Sec. 5.1) and to version the *Frontier Registry* with a domain k -target.

Appendix E: Toolkit and Reproducibility

The GEC Toolkit Pro implements the Generalized Efficiency Coefficient core (Python 3.11, NumPy 1.26). The CaseStudies notebook reproduces the results in Table 3 and Figs. 3–6. Both are available at github.com/project-tower/gec-toolkit-pro.

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